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Unveiling market dynamics: a machine and deep learning approach to Egyptian stock prediction

Ibrahim Eldesouky Fattoh[1], Marwa El Maghawry Ibrahim[2†] and Farid Ali Mousa[3*†]

Abstract

Accurate stock price forecasting is essential for making smart investing choices. In the context of the Egyptian stock market, this study examines the predictive capabilities of several machine learning and deep learning models for stock price prediction. Five different datasets with historical stock price information from significant Egyptian companies are used in the study methods such as Random Forest, Linear Regression, LSTM, and Bi-LSTM which were employed and evaluated using performance metrics including Mean Squared Error (MSE), Mean Absolute Error (MAE), and R-Squared. Among these, the Bi-LSTM model consistently delivered the most accurate price forecasts due to its superior ability to capture complex patterns and relationships inherent in stock market data. Additionally, statistical models ARIMA and GARCH were implemented for comparative purposes, but the machine learning and deep learning models outperformed these traditional approaches. This study contributes to a deeper understanding of stock price behavior in the Egyptian market, providing valuable insights for financial analysts and investors.

Keywords Stock price forecasting, LSTM, Bi-LSTM, Egyptian stock market

Introduction

Many individuals dedicate a significant portion of their time to the pursuit of wealth, making substantial sacrifices to secure a comfortable and stable existence for their families. One of the main alternatives to achieve long-term wealth is the investment is the stock market, as investing in the stock market is a long-term strategy that is expected to generate returns that exceed those of a savings account, particularly if we diversify our investments to mitigate risks. Despite its potential, a substantial portion of the population in a variety of countries is still hesitant to invest in the stock market, often perceiving it as too complex or hazardous. This anxiety is the result of a lack of comprehension, skepticism regarding the market's capacity to generate consistent profits, and the potential for loss. The aim of most traders is to buy stocks at cheap prices and subsequently resell them at a premium price. Predicting stock prices was from the most important issues for the investors and companies. From that, the primary motivation for this research stems from the need to improve the accuracy and reliability of stock price predictions in the Egyptian business environment. The Egyptian stock market, as an emerging market, presents unique characteristics and volatility patterns that are often overlooked in global research, and the increasing complexity of the global financial landscape has driven the demand for more sophisticated and objective analytical tools that can yield actionable insights.

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Traditional methods of predicting stock prices are used for a long period of time, and it is still useable. There are two traditional famous methods of stock analyses. The first method is fundamental analysis, in which they evaluate the intrinsic value of stocks, by evaluating the performance of the industry, economy, and companies' financial performance. Some research papers explored the fundamental analysis-based approach to stock market forecasting which entails the following techniques: calculating the weight of financial indicators, evaluating and selecting individual stocks, selecting financial news features, determining stock trading signals based on financial news, and predict stock price trends [1]. Some other researchers did the fundamental analysis using the economic condition of the company, its employees, board of directors, financial health, annual reports, balance sheets, income statements, as well as external factors such as natural or man-made disasters and political information to forecast future stock prices [2–6). Many results showed that fundamental analysis is effective for predicting long-term stock price movements but is not well-suited for short-term price fluctuations [7].

Alternatively, the second method is the technical analysis. This is a method of examining the evolution of equities by examining the statistics generated by market activity, such as past prices and volumes. Researchers have devised numerous patterns and signals to facilitate technical analysis trading. Additionally, technical analysts have devised a variety of trading systems to facilitate their forecasting and trading of price fluctuations. Marshall [8] studied the use of candlestick technical analysis and found no evidence to support its effectiveness in predicting future prices for the hundred stocks listed on the Tokyo Stock Exchange, deeming it unprofitable. In contrast, [9], [10] examined the effectiveness of technical analysis in forecasting future stock prices on the Amman Stock Exchange, concluding that technical analysis could indeed predict stock prices and provide profitable buy and sell recommendations. Some other technical indicators such as the simple moving average, the exponential moving average (EMA), and moving average convergence/divergence rules (MACD) are used in some studies as the study of [4] and the studies of Bisoi and Dash [11] and [12] to explore the benefits of technical analysis for stock price prediction, but the studies yielded mixed results.

The solution to this challenge was using the advanced machine learning techniques. The aim of using these advanced technologies is to provide accurate and reliable predictions that can be useful for even those who do not have any financial background. Artificial Intelligence (AI) become one of the most important newest advanced technologies. Artificial intelligence allows a machine to think like a human by mimicking the structure of the human brain. It is used in many fields, such as robotics, game theory, the study of cognition, and reasoning and machine learning [13] [14]. A typical machine learning model pipeline for predicting stock price consists of multiple phases: gathering historical data, preprocessing the data, developing a model for forecasting, and assessing the model. Preprocessing consists of deleting zero values, reducing duplication, and scaling features. Subsequently, significant traits are shortlisted, and legitimate data are chosen for stock price prediction [15] [16]. Although machine learning (ML) and deep learning (DL) models have gained traction as modern solutions to enhance forecasting accuracy, there remains a notable gap in applying these advanced techniques to specific, less-explored stock markets, such as the Egyptian Stock Exchange. The Egyptian stock market presents unique challenges and opportunities due to its market dynamics, economic conditions, and investor behaviors.

This research aims to bridge the gap by implementing various machine learning and deep learning algorithms, including Random Forest, Linear Regression, LSTM, and Bidirectional LSTM, to address stock price prediction for different companies on the Egyptian Stock Exchange. The goal is to achieve accurate results, anticipate market movements, and capitalize on emerging trends, thereby contributing to the understanding of how these advanced models perform in non-mainstream markets and potentially providing practical solutions for investors with limited financial expertise. In addition, the integration of machine learning approaches within the Egyptian business environment aligns with global trends in financial technology (FinTech), ensuring that the market remains competitive and adaptive to modern investment practices. This progress supports the broader goal of advancing financial innovation in Egypt, reinforcing its position in regional and global markets, and contributing to economic resilience through more informed investment strategies.

The structure of this paper is as follows: Sect. 2 reviews the existing literature on the topic. Section 3 presents a detailed description of the dataset and the methodologies employed. In Sect. 4, we introduce and describe the proposed model. While Sect. 5 introduces the results. Section 6 offers an analysis and discussion of the results. Finally, Sect. 7 concludes the paper and outlines potential directions for future research.

Literature review

Some research has been done on the benefits of using machine learning in predicting the stock prices. Reddy [14] study employed the machine learning approach called Support Vector Machine (SVM) to predict stock

prices for large and small capitalizations in three separate marketplaces, the study used both daily and up-to-the-minute price data. The study model yields greater profitability in comparison with the chosen benchmarks.

Aldin et al. [17] conducted an assessment of predicting the future direction of the stock price index of the Tehran Exchange, their study primarily focused on evaluating the effectiveness of practical indicators such as Moving Average, RSI, CCI, MACD, etc., in relation to elements such as stock price, stock volume, and stock interest rate. The data comprised the overall stock price index, which consisted of the closing, high, and low prices. A three-layered artificial neural network (ANN) model based on a feed structure has been utilized for the prediction of stock prices. After performing calculations, the findings were evaluated using validation techniques such as the index Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE), and Root-Mean-Square Error (RMSE), and the displayed data exhibit a forecasted accuracy rate of 94%.

Bonde et al. [18] tried to assess the movement direction of stock prices, they analyzed eight different stocks to assess whether the prices will increase or will drop. They employed genetic alogorithms (GA-ANN) to determine the relationship weight for each attribute, to enable accurate predictions of stock prices. They gathered data about eight stocks over 5 years: Google, Apple, IBM, Abode, Microsoft, Oracle, Sony, and Symantec. The results showed that genetic algorithm achieved an accuracy of 73.87%, while the evolutionary strategies achieved an accuracy of 71.77%. Madge and Bhatt [19] developed an approach to predict the prices of 34 technical stocks. The study employed the Support Vector Machine (SVM) to achieve its objective which was to determine the degree of price irregularity and momentum for each stock during the period from 2007 till 2014. It was found that the SVM may not predict the price for the next day, and it has a strong predictive capacity in the long term with a predictive value of 60% or higher.

Wanjawa [20] conducted an analysis to determine the stock prices listed in Shanghai Stock Exchange, his aim was to forecast the stock prices for seven stocks over a 15-day period from September 21, 2016, to October 11, 2016, without relying on the actual values. The study results demonstrated that basic neural network systems may effectively forecast stocks with high precision.

Another study by Zeng and Liu [21] who performed an assessment of the volatility and difficulties encountered in the stock market. They aimed to analyze the learning and fitting performance of various stock data. Their predictive model was based on Long Short-Term Memory (LSTM) and Neural Network (NN). After completing the comparative analysis, it was shown that the LSTM outperforms the NN in terms of scoring. The average accuracy of LSTM is 12 percentage points higher than that of NN. LSTM outperforms other models in accurately predicting weekly data over multiple days.

In order to address the unpredictable and ever-changing character of the market, Sheth and Shah [18] established a connection between the stock market and artificial intelligence, resulting in remarkable outcomes. They implemented the prediction technique using three methods: Artificial Neural Network (ANN), Support Vector Machine (SVM), and Long Short-Term Memory (LSTM). ANN utilizes neural networks, SVM employs the Kernel approach, and LSTM utilizes the Keras LSTM framework. The analysis determined that the ANN method, which is based on neural networks, yields the most optimal results. This is due to its ability to evaluate intricate, nonlinear relationships and effectively identify patterns. Their results also showed that SVM is a relatively new method and has the potential to yield improved results in the future. On the other hand, LSTM performs well only when provided with a large dataset, which might be considered a limitation. In their study, Han and Fu [22] proposed a bidirectional long short-term memory (Bi-LSTM) model to forecast future stock prices based on historical data. This model extends the capabilities of the standard LSTM by processing input sequences in both forward and backward directions, enabling it to capture both short- and long-term dependencies in stock price data. Han and Fu applied the Bi-LSTM model to Apple Inc.'s historical stock price data and evaluated its performance using mean squared error (MSE) and visual comparisons of predicted versus actual prices. Their findings revealed that the Bi-LSTM model performed well in making accurate predictions and identifying trends, although it struggled with abrupt market shifts. The results underscored the potential of the Bi-LSTM model as a promising tool for stock price forecasting and broader financial applications. Ma et al. [23] explored in their study the limitations of traditional stock price prediction approaches that predominantly rely on numerical features and sentiment analysis of news related to target stocks. They proposed the Multi-source Aggregated Classification (MAC) method, which integrates numerical features, market-driven news sentiments of target stocks, and the news sentiments of related companies. To enhance the representation of real market sentiments, they developed an embedding feature generator pre-trained to align news data with actual stock price movements, enabling sentiment representation in a vector space. Additionally, MAC incorporates a graph convolutional network to capture the interconnected effects of related companies on the target stock. Their experimental results demonstrated that MAC significantly

outperforms existing methods in predicting stock price movements, achieving superior performance in metrics such as the Sharpe Ratio and back testing trading income. Fan et al. [24] proposed in their research, an advanced stock price prediction model using the Extended Long Short-Term Memory (xLSTM) algorithm, which aims to improve predictive accuracy over both short- and long-term periods. Through extensive experiments involving models based on xLSTM and standard LSTM architectures across various stocks, they found that the xLSTM consistently outperformed the LSTM. The performance gap was particularly pronounced as the prediction period lengthened, indicating that the xLSTM model is more adept at capturing long-term patterns in financial data. This work presents a promising solution for enhancing the accuracy and reliability of stock price predictions. There is a significant shortage of recent literature on the application of machine learning for stock price prediction. The existing literature, while demonstrating notable advancements, remains limited in several key areas. Prior studies exhibited constraints in predictive accuracy, model generalizability across diverse market conditions, and the capacity to effectively capture both short-term fluctuations and long-term dependencies within financial data. These limitations suggest that further research is needed to develop more comprehensive and robust machine learning models capable of addressing the complexities inherent in stock market behavior.

Materials and methods

To forecast stock prices and market trends in the Egyptian market, this study employs a comprehensive methodology that aims to provide a wide range of solutions based on machine learning and deep learning techniques. Data collection, preprocessing, model selection, and performance evaluation are some of the steps in the methodology. To identify both linear and intricate nonlinear patterns in the stock data, we combined cutting-edge deep learning architectures (such as LSTM and Bi-LSTM) with conventional machine learning models (such as random forest and linear regression). The datasets used are described first in this section, and then, each machine learning and deep learning technique used, along with its settings, is thoroughly explained.

Dataset

The researchers used stock market data for the most five traded companies in November 2023 as the dataset for this analysis. The dataset was downloaded from ivesting.com. The companies are listed in two different sectors: Financial Services and Construction: (a) Belton Financial Holding (BTFH) (b) Madinet Nasr (MASR) (c) Orascom Investment (OIH) (d) Palm Hills Development (PHDC) e) Qalaa Holdings (CCAP). The data are daily from 1-10-2013 to 22-11-2024. So each company has around 2460 rows and 8 columns. Each row represents the information of a single day and for the columns:

1st column—the row number
2nd column—the daily date
3rd column—the closing price of that day
4th column—the opening price of that day
5th column—the highest price of that day
6th column—the lowest price of that day
7th column—the trading volume of that day and
8th column—the percentage change.

Methods

Linear regression

A linear regression uses labeled data (supervised learning) to generate a relationship among variables that are independent and dependent using basic mathematical formulas, resulting in an optimal fit line or line of least reluctance. The resulting line can be used for making stock forecasts through graph or curve analytics [25].

Linear regression is an approach for predicting scalar outputs given one or more explanatory inputs. The relationship is generated by linear predictor functions, with all parameters changed and learned in data.

$$Y_i = \beta_0 + \beta_1 X_i. \tag{1}$$

Initially, the values of β_0 and β_1 are set at random. The model adjusts β_0 and β_1 values based on the X-Y line's fit to the data curve. Specifically, all of the variables and parameters in the equation may be vectors. Equation (1) is used to design a straight line that passes through as many data points as possible. To chart dataset numbers on a chart, a straight line is mathematically matched to minimize the square of the distance that exists between each point and the line.

Random forest

One method of ensemble machine learning is Random Forest (RF). Both classification and regression tasks can be completed by it. Instead of depending solely on individual decision trees to reduce the variance in the model, the aim is to aggregate numerous decision trees to decide the final result. According to Vijh et al. [26], each decision tree in this study is trained using newly formed variables, which, in turn, affect the decision at the tree's nodes. At each node, a random selection of attributes is used to build the individual decision tree classifier models. In order to add as much randomization as possible, the feature and feature value are chosen at random for splitting at each node. Avoiding overfitting of the models during the training phase is the goal of adding a significant amount of randomness to the decision tree model

construction process. Each tree casts a vote during the classification stage, and the test case is given the class that most of the trees return.

LSTM

An unsupervised learning model called LSTM is a type of Recurrent Neural Network (RNN) that may be used to classify, assess, and predict time-series data as well as learn order dependency in chronology prediction issues. The ability of long short-term memory to manage the entire set of inputs gives it a major edge over other common neural networks. This is how LSTMs are designed to circumvent the problem of long-term dependency. It can handle the entire dataset and has a larger memory than other neural networks. The forget, input, and output gates stand in for the cells that comprise lengthy short-term memory. The information flowing into and out of the cell is synchronized by the three gates. Keeping track of the relationships between the components in the input sequence is made easy by the cell. The forget gate helps the model eliminate information that is not needed. Data are incorporated into the cell's state with the help of the input gate. The output gate chooses and stores the data that will be displayed [27] [26]. In order to estimate the price values for the various firms, the LSTM model was adjusted to include historical stock data that were gathered using Python modules.

Bidirectional LSTM

The Bi-LSTM is a recurrent neural network (RNN) that is ideal for analyzing time-series data. The Bi-LSTM is a version of the Long Short-Term Memory (LSTM) model that can handle sequences of input in both the forward and backward orientations [22]. The Bi-LSTM architecture builds on the original LSTM by including an additional LSTM layer which processes the input data sequence in the opposite order. This enables the model to capture both forward and backward correlations in its input data, making it valuable for applications such as natural language processing, speech recognition, and forecasting stock prices [28]. The Bi-LSTM model has an advantage over the conventional LSTM in that it can more accurately represent dependency over time in the input data sequence. This is due to the backward LSTM layer can reveal insight about the sequence's finish, which can be used to forecast future values. To apply the Bi-LSTM model for stock price prediction, we first normalize and transform the input data into fixed-length sequences. To train the Bi-LSTM model, we partition the data into training and testing sets and use the mean squared error (MSE) loss function. After training, we may run the model on the test data in order to make predictions about future values.

Proposed approach

The three primary stages of the suggested model for stock price prediction—data preprocessing, model training, and prediction—are depicted in Fig. 1. Every stage is essential to guaranteeing the model's overall efficacy and dependability. Data Preprocessing Phase: To get the raw data ready for analysis, the preprocessing phase is crucial. Several crucial steps are included in this phase:

- *Handling Missing Values* Missing data points are managed using imputation techniques to maintain data integrity without introducing biases.
- *Data Cleaning* Procedures are implemented to remove errors, inconsistencies, and outliers, ensuring that the dataset is reliable and accurate.
- *Normalization* Data normalization is performed to scale the features, allowing all variables to contribute equally to the training process. This step is crucial for deep learning models that are sensitive to scale differences.

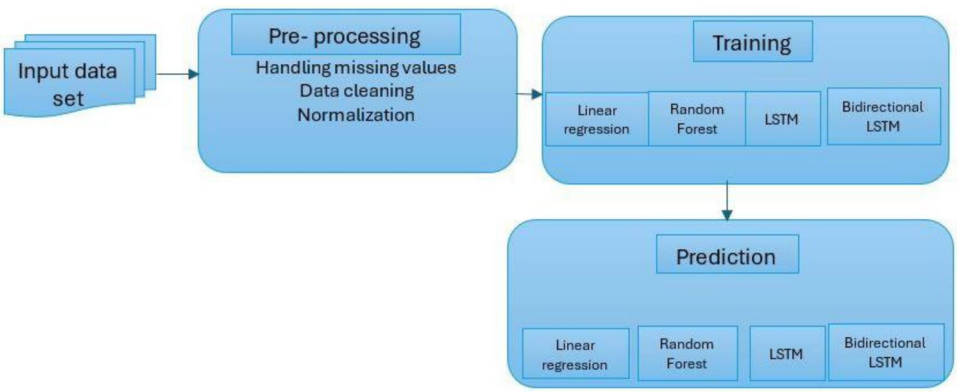


Fig. 1 Proposed approach

This stage is crucial because it converts unprocessed data into a format that can be used for model training, which enhances the models' performance and learning process. The preprocessed data are divided into training and validation sets throughout the training phase. The training dataset is used to train four models: LSTM, Bidirectional LSTM, Random Forest, and Linear Regression. This stage guarantees that every model gains the ability to forecast and discover the underlying patterns in the data. This phase's significance stems from its capacity to evaluate the performance of multiple models and determine which one has the best forecast accuracy. Prediction Phase: Once trained, each model is used to predict future stock prices based on new, unseen data. The models' outputs are compared to evaluate their accuracy, providing insights into the best-performing method for stock price prediction.

Results and discussion

In this section, the results obtained from the experiments applied on the five different companies are presented. The four models described before were implemented and evaluated against each dataset mentioned before. To evaluate the different models, three metric values were used which are Mean Square Error (MSE), Mean Absolute Error (MAE), and R-Squared (R^2) values.

Mean Square Error (MSE) measures the average of the squares of the errors, where the error is the difference between the actual value and the predicted value. MSE can be calculated using Eq. 2

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - y_i'\right)^2 \tag{2}$$

where n is the number of data points, yi is the actual value, and y'I is the predicted value. It is selected because it penalizes larger errors more severely, providing a clear measure of how well the model's predictions align with actual values. It is valuable in this context as it helps identify models that minimize significant prediction errors.

Mean Absolute Error (MAE) chosen for its simplicity and interpretability as it measures the average magnitude of the errors in a set of predictions, without considering their direction. It provides a linear score that treats all errors equally. Also it is less sensitive to outliers than MSE since it does not square the errors. MAE can be calculated using Eq. 3

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - y_i'\right| \tag{3}$$

R-Squared value (R2) known also as the coefficient of determination, used to assess how well the model explains the variability in the data. It measures the proportion of the variance in the dependent variable that is predictable from the independent variable(s). R2 can be calculated using Eq. 4

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(y_i - y_i'\right)^2}{\sum_{i=1}^{n}\left(y_i - \bar{y}\right)^2} \tag{4}$$

The results obtained for stock price prediction for the different companies are summarized in Table 1.

Table 1 Price forecasting for different models with different datasets

Company	Metric	Linear Regression	Random Forest	LSTM	Bi-LSTM
MASR	MSE	0.0005	0.0009	0.0005	0.0005
	MAE	0.015	0.0215	0.0167	0.0100
	R-Squared	0.9997	0.9992	0.9996	0.9997
BTFH	MSE	0.0231	0.0753	0.0244	0.010
	MAE	0.063	0.0667	0.0558	0.059
	R-Squared	0.9984	0.9949	0.9983	0.999
CCAP	MSE	0.0002	0.0006	0.0004	0.0002
	MAE	0.0091	0.0146	0.0140	0.0103
	R-Squared	0.9998	0.9994	0.9996	0.9999
OIH	MSE	0.00004	0.00013	0.0004	0.00003
	MAE	0.00307	0.0044	0.00303	0.0048
	R-Squared	0.9996	0.9988	0.9996	0.9994
PHDC	MSE	0.00023	0.00049	0.00029	0.00023
	MAE	0.01058	0.01556	0.0122	0.01011
	R-Squared	0.9997	0.9993	0.99964	0.9997

Table 1 shows the obtained results for the price forecasting attribute from the four implemented models for the five different datasets collected in this study.

Figures 2, 3, 4, 5, and 6 show the stock price prediction for the five datasets.

The four subplots (a, b, c, and d) in each of Figs. 2 through 6 show the comparison of the actual stock price data with the anticipated values derived from four distinct models: LSTM, Bi-LSTM, Random Forest, and Linear Regression as an overview of the

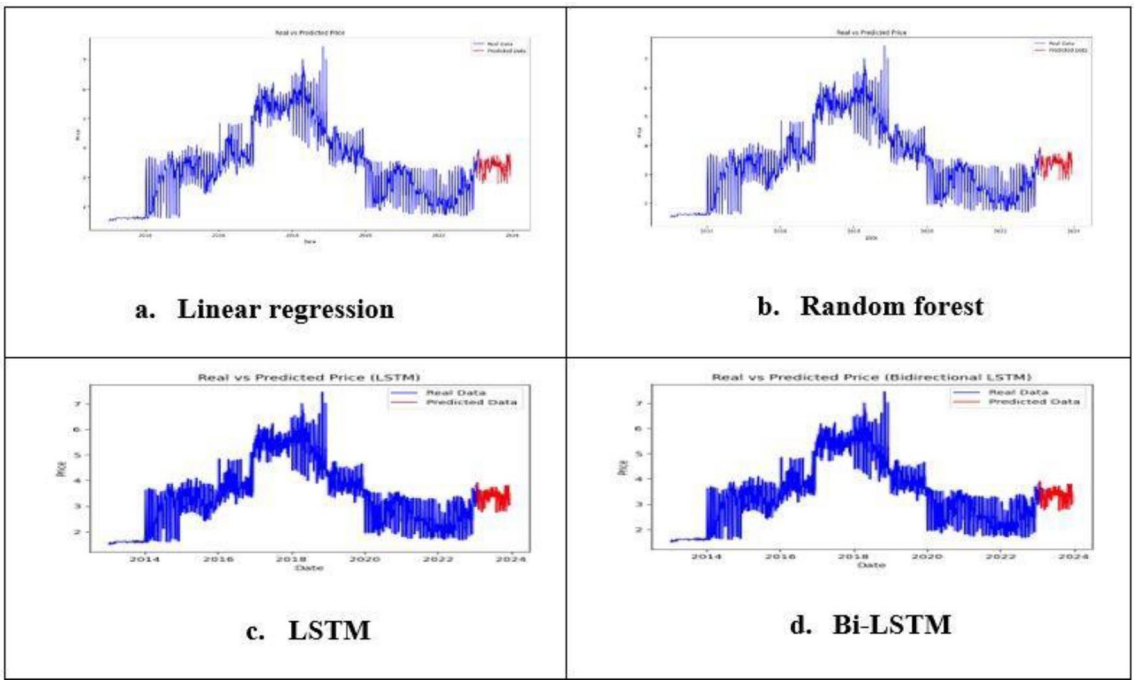


Fig. 2 MASR company stock price forecasting

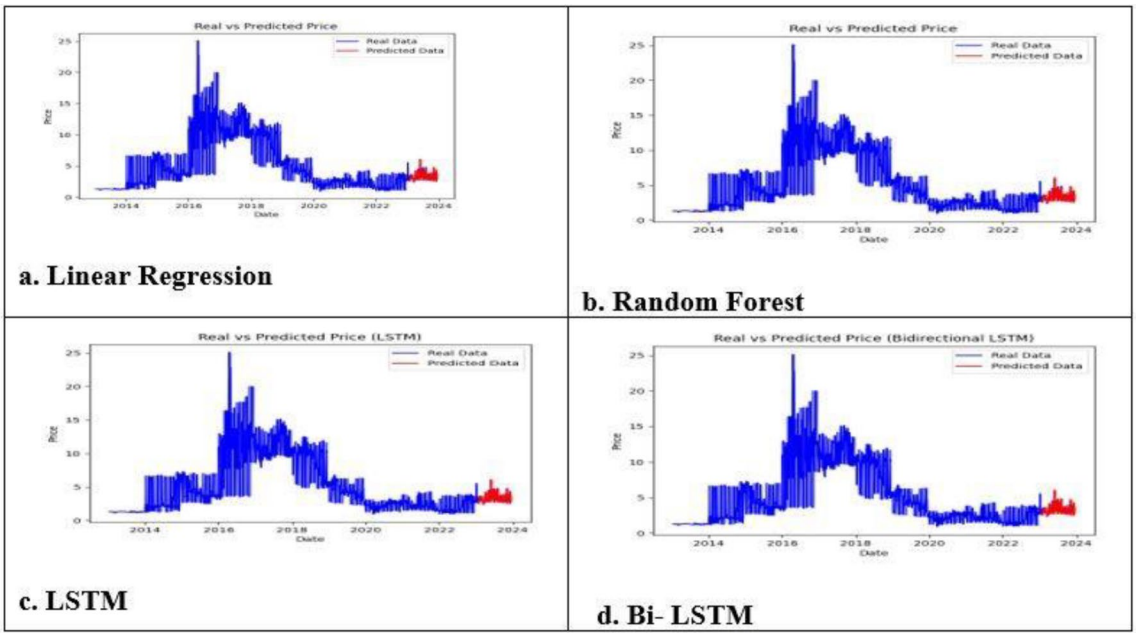


Fig. 3 BTFH company stock price forecasting

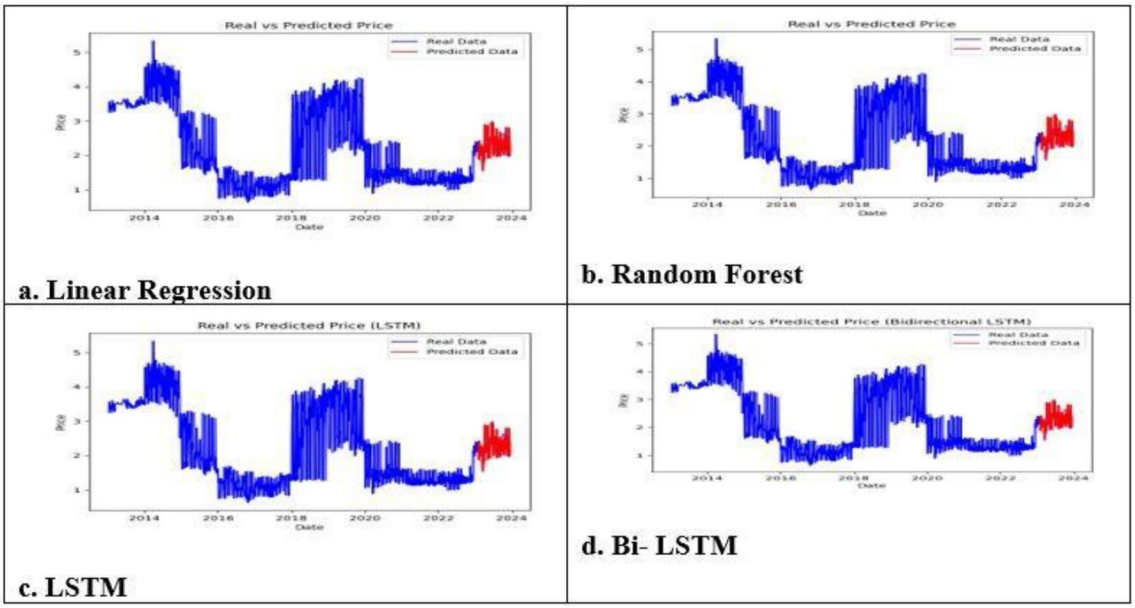


Fig. 4 CCAP company stock price forecasting

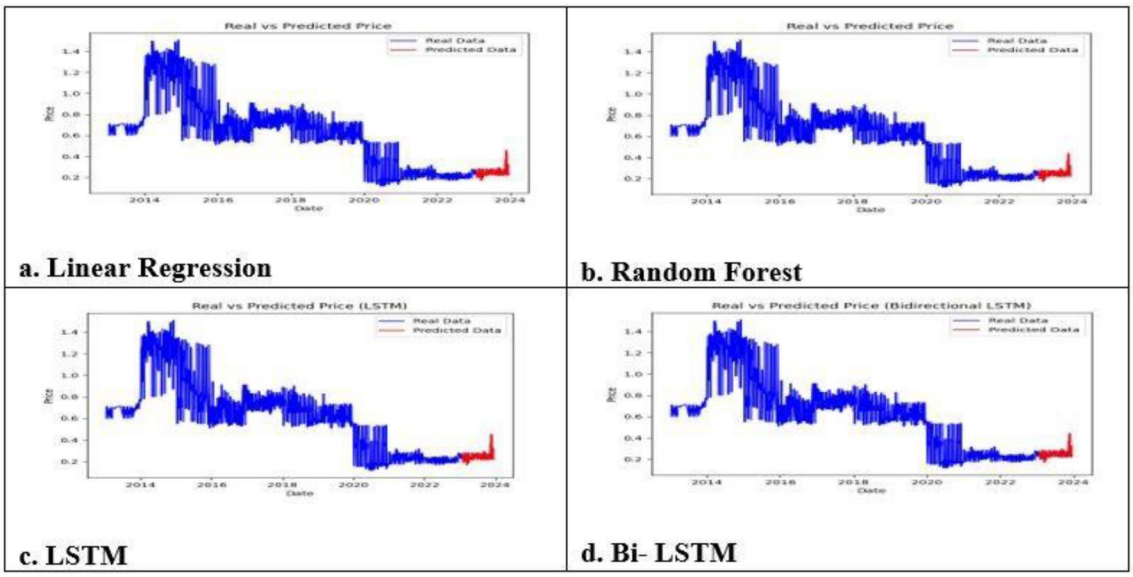


Fig. 5 OIH company stock price forecasting

figures mentioned above. Although it may capture broad trends, linear regression has trouble with intricate patterns, which leads to significant departures from the actual stock values. Compared to linear regression, Random Forest performs better at handling nonlinear correlations, although it is still unable to adequately represent abrupt spikes and troughs. Long Short-Term Memory, or LSTM, makes predictions that are more accurate with time by identifying nonlinear dynamics and sequential dependencies. Bi-LSTM (Bidirectional LSTM): Outperforms all other models by effectively capturing intricate patterns and dependencies in the data, resulting in the closest match to the real stock prices, especially during periods of significant fluctuations.

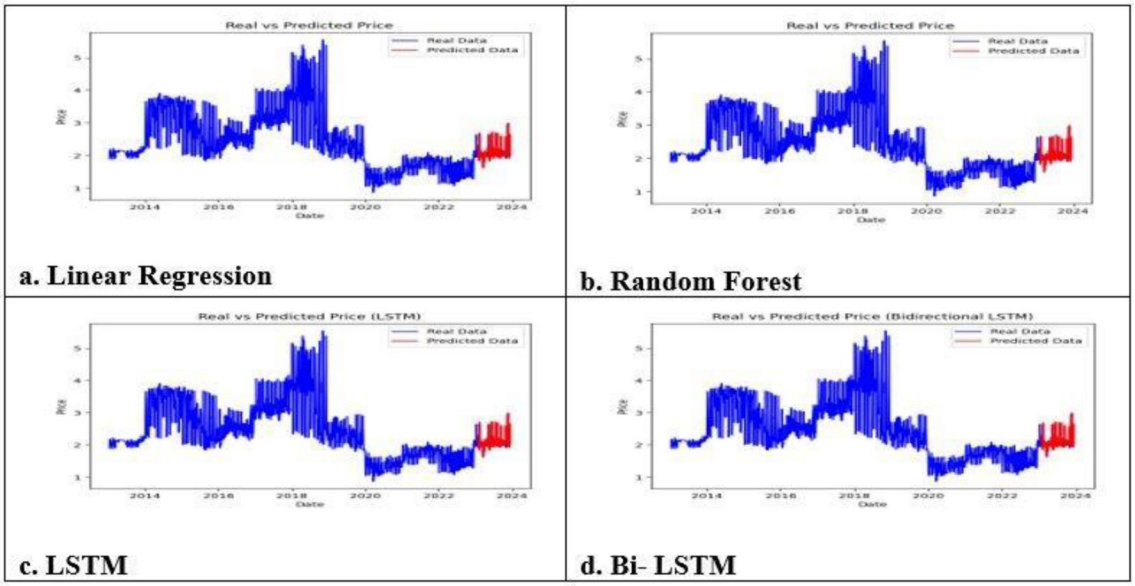


Fig. 6 PHDC company stock price forecasting

Analysis of stock price forecasting results

The results in Fig. 7 indicate a significant variation in performance across the different models and datasets. Bi-LSTM and linear regression models generally exhibit lower MSE values, suggesting better predictive accuracy compared to Random Forest and LSTM. The dataset labeled BTFH shows the highest MSE values across all models, indicating potential challenges in forecasting for this particular dataset. Further investigation into the characteristics of the BTFH dataset might be necessary to understand these performance disparities. Overall, Bidirectional LSTM stands out as the most effective model for price forecasting in this analysis, followed by Linear Regression with certain reservations, and then Random Forest and LSTM as viable alternatives with respect to MSE.

The following Fig. 8 presents a comparison of Mean Absolute Error (MAE) values for different forecasting models. MAE is a measure of error, indicating the average magnitude of the prediction errors. Lower MAE values signify better model performance. It can be noticed that Bi-LSTM and linear regression models

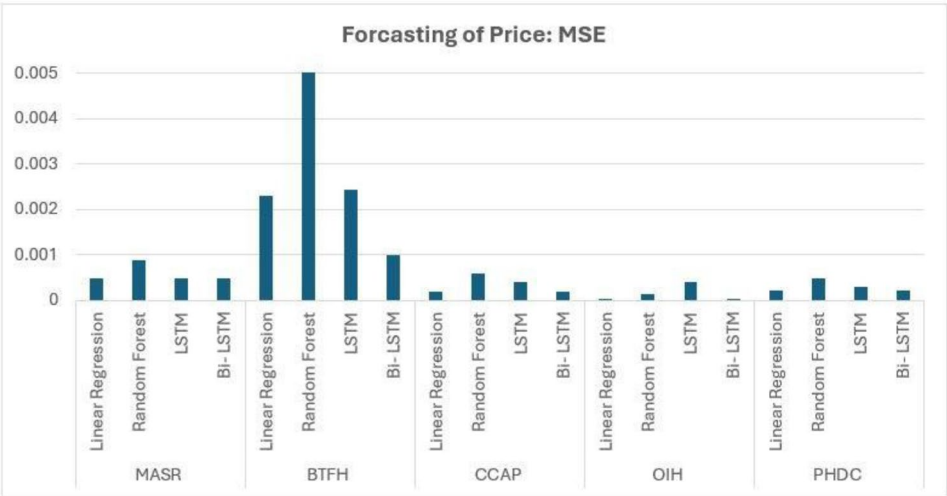


Fig. 7 MSE values for price forecasting

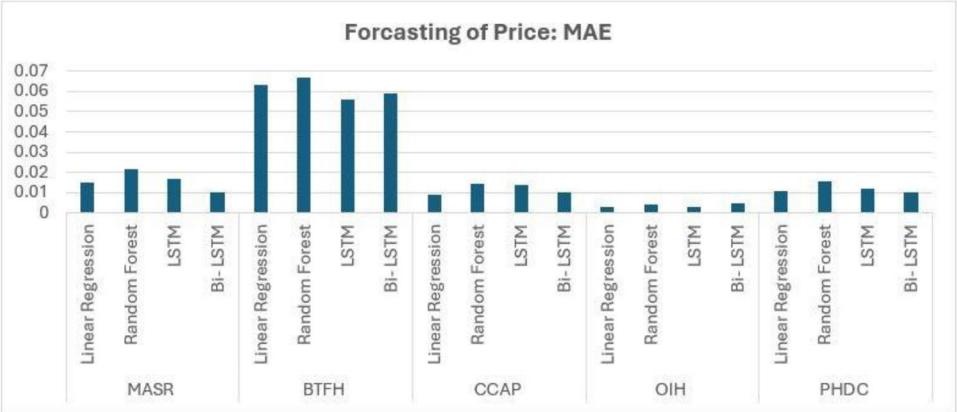


Fig. 8 MAE values for price forecasting

generally exhibit lower MAE values across most datasets, suggesting superior predictive accuracy compared to Random Forest and LSTM. Also the dataset significantly influences model performance. It is shown that BTFH also presents the most challenges for all models, as evidenced by consistently higher MAE values. Overall, this chart reinforces the findings from the previous MSE chart, strengthening the evidence for the superior performance of Bi-LSTM and linear regression models in terms of minimizing prediction errors.

Figure 9 presents a comparison of R-Squared values for different forecasting models. It can be observed that Bi-LSTM and linear regression models consistently demonstrate higher R-Squared values across most datasets, suggesting a stronger explanatory power for these models in predicting price. Overall, the R-Squared chart strongly supports the findings from the previous MSE and MAE charts, solidifying the conclusion that Bi-LSTM then the linear regression and is superior models for price forecasting in this context.

From the previous results obtained, it is shown that the best prediction results in price prediction were Bi-LSTM model as a result of its ability of capturing long-term dependencies in the data which enables the model to better understand the overall trend and patterns in the data and its powerful feature extractor, which automatically learning relevant features from the input data.

To ensure the power of the deep learning and machine learning models implemented, a two famous statistical models for prediction were implemented. The two models are Autoregressive Integrated Moving Average (ARIMA) [29] and Generalized Autoregressive Conditional Heteroskedasticity model (GARCH) [30]. The obtained results of the two models were compared with the early provided machine and deep learning models, and the obtained results for stock price prediction are shown in Table 2.

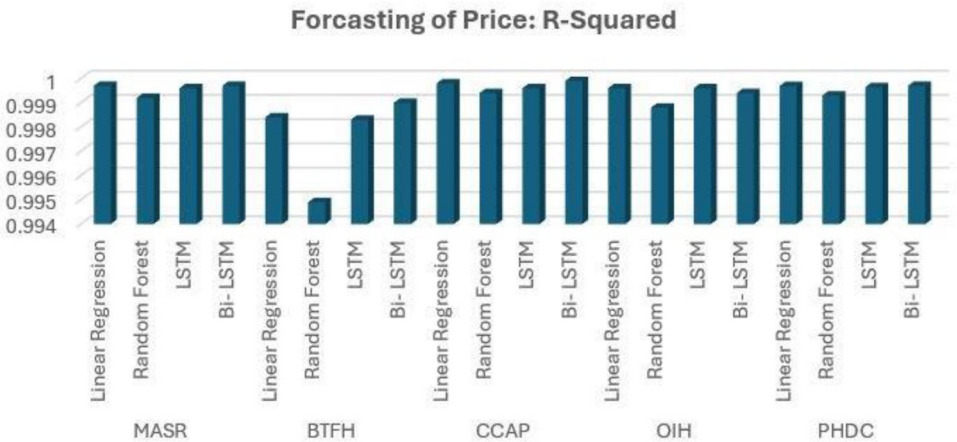


Fig. 9 R-Squared values for price forecasting

Table 2 Comparing the stock price forecasting of the ML and DL models with ARIMA and GARCH models

Company	Metric	Linear Regression	Random Forest	LSTM	Arima	GARCH	Bi-LSTM
MASR	MSE	0.0005	0.0009	0.0005	0.405	0.6831	0.0005
	MAE	0.015	0.0215	0.0167	0.574	0.6308	0.0100
	R-Squared	0.9997	0.9992	0.9996	0.001	0.7434	0.9997
BTFH	MSE	0.0231	0.0753	0.0244	1.217	1.0393	0.010
	MAE	0.063	0.0667	0.0558	0.98	0.8877	0.059
	R-Squared	0.9984	0.9949	0.9983	0.124	0.0209	0.999
CCAP	MSE	0.0002	0.0006	0.0004	0.46	0.474	0.0002
	MAE	0.0091	0.0146	0.0140	0.489	0.4977	0.0103
	R-Squared	0.9998	0.9994	0.9996	0.63	0.745	0.9999
OIH	MSE	0.00004	0.00013	0.0004	0.0032	0.1961	0.00003
	MAE	0.00307	0.0044	0.00303	0.0498	0.4411	0.0048
	R-Squared	0.9996	0.9988	0.9996	0.28	0.3687	0.9994
PHDC	MSE	0.00023	0.00049	0.00029	0.14	0.4757	0.00023
	MAE	0.01058	0.01556	0.0122	0.306	0.5822	0.01011
	R-Squared	0.99969	0.9993	0.99964	0.03	0.7752	0.9997

From the results obtained in Table 2, it is shown also that the Bi-LSTM is still performing better the two statistical models and nearly all the provide machine and deep learning model performs better than this two statistical models. Highlighting the limitations of the statistical models in capturing the complex dynamics of stock price movements in emerging markets such as Egypt. These findings underscore the potential of deep learning techniques, particularly Bi-LSTM, in financial forecasting applications.

Conclusion and Future Work

This study examined the use of machine learning, deep learning, and statistical models to the prediction of stock prices in the Egyptian market. Our experiments demonstrated that the Bi-LSTM model continuously outperformed other models in terms of prediction accuracy, as indicated by MSE, MAE, and R-Squared values. Bi-LSTM's remarkable performance can be attributed to its ability to identify complex, nonlinear relationships, and long-term dependencies in financial time-series data. However, the poor performance of the GARCH and ARIMA models suggests that traditional statistical methods may not be the greatest fit for stock price prediction given the volatile and unpredictable nature of developing markets. The study offers empirical proof that complex and volatile market data need the use of advanced model architectures, especially in developing economies.

Practical Implications: This study emphasizes how important it is to use cutting-edge deep learning techniques for financial forecasting, especially in unstable markets like Egypt's. Investors and financial analysts can use Bi-LSTM's excellent performance to improve portfolio management techniques, lower risks, and streamline decision-making processes. More accurate market insights and prediction skills may be provided by incorporating these discoveries into financial software, and Principal Limits: It is important to recognize the limits of this study, even in light of the encouraging findings. Only five Egyptian companies' data were used to train and test the algorithms, which would have limited how broadly the results can be applied. Furthermore, the robustness of forecasts in real-world scenarios may be impacted by the absence of explicit modeling of external elements such as market rules and unforeseen geopolitical events. Future Scope: To improve prediction accuracy even more, future studies could expand on this work by investigating the incorporation of extra input elements such sentiment analysis from social media, geopolitical events, and macroeconomic indicators. Furthermore, testing ensemble methods that incorporate several models may increase the robustness of predictions. The creation of domain-specific hybrid models that combine statistical and deep learning techniques, leveraging each of their advantages, is another exciting direction. Finally, more thorough insights into these models' efficacy and adaptability in various economic circumstances would be obtained by expanding their applicability to a wider range of businesses and industries in Egypt or other developing economies.

Abbreviations

AI	Artificial Intelligence
ARIMA	Autoregressive Integrated Moving Average
Bi-LSTM	Bidirectional Long Short-Term Memory
BTFH	Belton Financial Holding
CCAP	Qalaa Holdings
GA-ANN	Genetic algorithms

GARCH	Generalized AutoRegressive Conditional Heteroskedasticity
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MASR	Madinet Nasr
MSE	Mean Squared Error
OIH	Orascom Investment
PHDC	Palm Hills Development
RF	Random Forest
RNN	Recurrent Neural Network
R-Squared	Coefficient of Determination
SVM	Support Vector Machine

Author's contributions
The authors contributed equally to this work by analyzing the data, preprocessing, design, and implementing the machine learning/deep learning/statistical models. Finally; all authors read and approved the final manuscript.

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Availability of data and materials
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Declarations

Competing interests
The authors declare that they have no competing interests.

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